Machine Learning-Based Network Intrusion Detection System Using Python

Student Details

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Project Overview

This project builds a Machine Learning-based Network Intrusion Detection System (NIDS) that classifies network connections as either **normal** or **attack** traffic. It uses the NSL-KDD dataset and compares Logistic Regression and Random Forest classification models.

Dataset

- **Dataset:** NSL-KDD Network Intrusion Detection Dataset
- **Dataset link:** https://www.kaggle.com/datasets/hassan06/nslkdd
- **Training file:** `KDDTrain+.txt`
- **Testing file:** `KDDTest+.txt`
- **Original source:** https://www.unb.ca/cic/datasets/nsl.html

Project Description

The system preprocesses network-traffic records, encodes categorical features, scales numerical features, and trains machine-learning models to detect intrusions. The target variable is converted into two classes:

- `0` — Normal connection
- `1` — Attack connection

The project evaluates both models using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve.

Technologies Used

- Python 3
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Project Files

File	Description
`SurekaMJ_MachineLearningBasedNetworkIntrusionDetectionSystem.ipynb`	Complete project code
`requirements.txt`	Required Python libraries
`SurekaMJ_ProjectReport.docx`	Complete project report
`README.md`	Project overview and setup instructions

Setup Instructions

1. Install Python 3.9 or above.
2. Install the required libraries:

```
pip install -r requirements.txt
```

3. Download `KDDTrain+.txt` and `KDDTest+.txt` from the NSL-KDD Kaggle dataset.
4. Place both dataset files in the same folder as the Jupyter Notebook.
5. Open and run the notebook:

```
jupyter notebook SurekaMJ_MachineLearningBasedNetworkIntrusionDetectionSystem.ipynb
```

6. Run all cells from top to bottom.

Expected Output

The notebook displays:

- Dataset shape and missing-value check
- Class distribution of normal and attack traffic
- Training of Logistic Regression and Random Forest models
- Accuracy, precision, recall, and F1-score
- Confusion matrices and ROC curves

Key Results

Random Forest is expected to provide stronger intrusion-detection performance than Logistic Regression because it can capture non-linear relationships among network-traffic features. Recall is particularly important because failing to detect an attack can lead to a security breach.

Author

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